
Measuring AI Ability to Complete Long Time Horizon Doubling Every ~7 Months ×20 Software Tasks

overall capability
day doubling time

7

METR — Model Evaluation & Threat Research | March 2025

$$R^2=0.97$$

2019 → 2025

Exponential Growth in AI Capability

arxiv.org/abs/2503.14499

A New Metric: 50% Task-Completion Time Horizon

1

Assemble Task Suite

170 diverse software & research tasks spanning 1 sec to 8 hours of human effort

2

Measure Human & AI Performance

Record how long skilled humans take. Run AI agents. Collect 800+ human baselines (2,529 hrs total)

3

Fit Logistic Model → 50% Threshold

Find the task duration at which an AI agent succeeds 50% of the time — the time horizon

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Figure 2: Methodology overview — METR (2025)

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

HCAST 97 tasks

Diverse software tasks | 1 min–30 hrs

RE-Bench 7 tasks

ML research engineering | 8 hrs each

SWAA 66 tasks

Short atomic actions | 1–30 seconds

Example Tasks by Difficulty

Task Name	Human Time	Description
find_shell_script	3 seconds	Multiple choice: Which file is a shell script?
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix bug in molecular dynamics simulation input files
munge_data	56 minutes	Write Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30× speedup of backtesting tool

AI Task Horizon: 2 Seconds (GPT-2) to ~1 Hour 50 Min (o3) in 6 Years

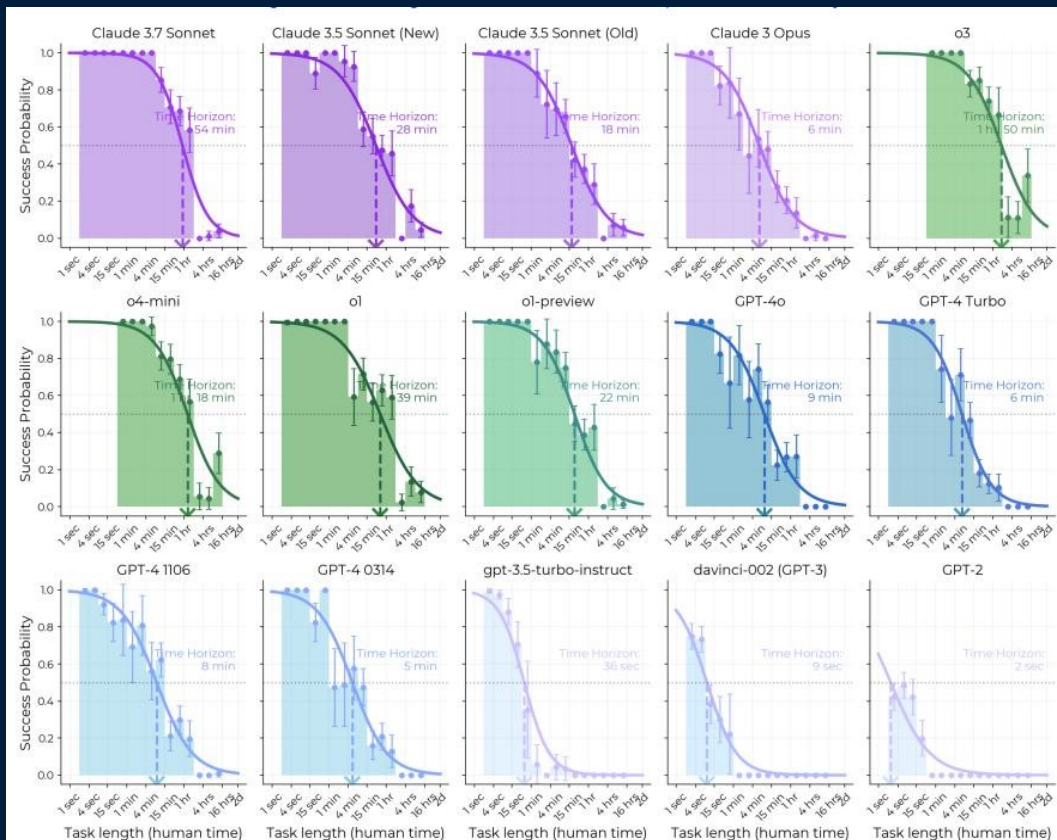


Figure 1: 50% task completion time horizon vs. model release date

~207 days

Doubling Time

$R^2 = 0.97$ | 2019-2025

Key Model Milestones

GPT-2 (2019) ~2 seconds

GPT-4 (2023) ~5 minutes

GPT-4o (2024) ~9 minutes

Claude 3.5 Sonnet (2024) ~28 min

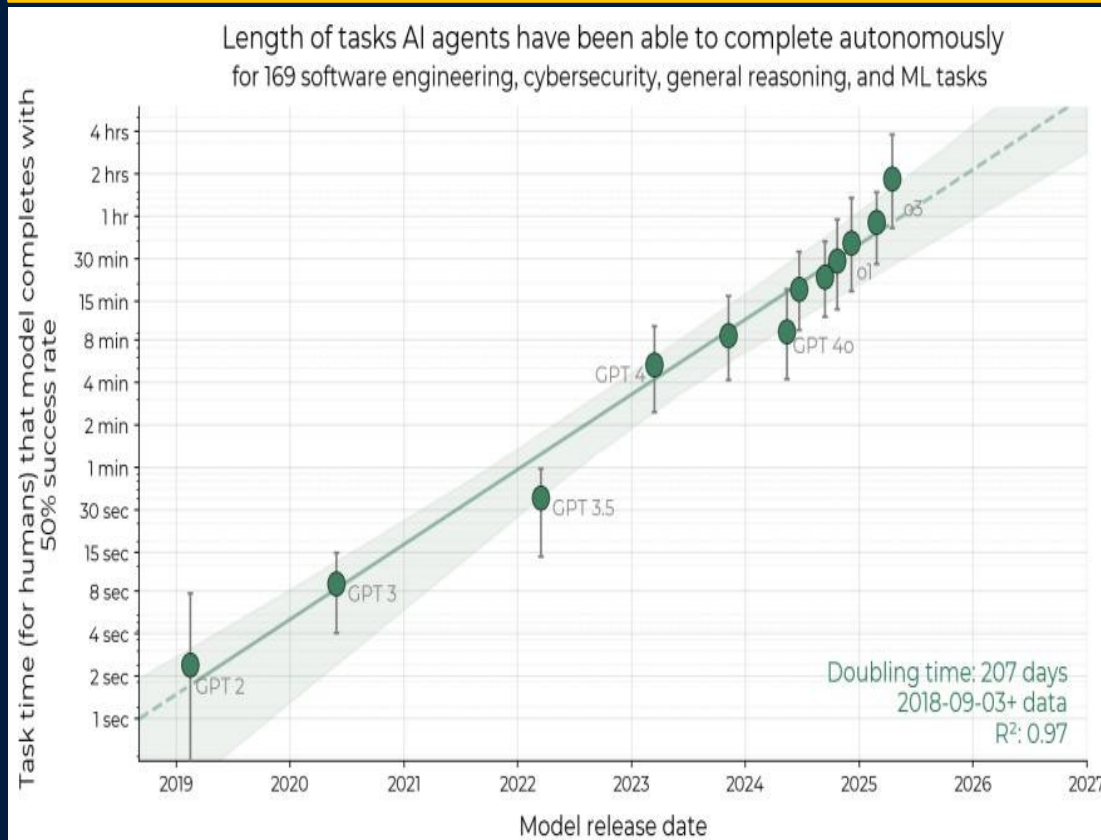
o1 (2024) ~39 minutes

Claude 3.7 (2025) ~54 min

o3 (2025) ~1 hr 50 min

2023-2025 growth ~20% faster than overall

Success Probability Drops Sharply as Task Length Increases



Key Observations

Frontier Shift Rightward

o3 (1h50m) vs GPT-2 (2 sec)

Sharp Decline Pattern

All models show steep drop-off at their time horizon

80% Horizon $\approx 5\times$ Shorter

80% success threshold is $\sim 5\times$ shorter than 50% horizon

Consistent Curve Shape

Logistic model fits well across all eras

o3: ~ 1 hr 50 min vs GPT-2: ~ 2 sec

Figure 3: Per-model success probability curves — METR (2025)

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

1

Improved Logical Reasoning

Models solve multi-step problems and maintain context across longer task chains

2

Better Tool Use

Improved use of bash, file I/O, code execution, and external APIs during autonomous execution

3

Reliability & Self-Awareness

Greater ability to verify own outputs, detect errors, and avoid compounding mistakes over long tasks

4

Adaptive Error Recovery

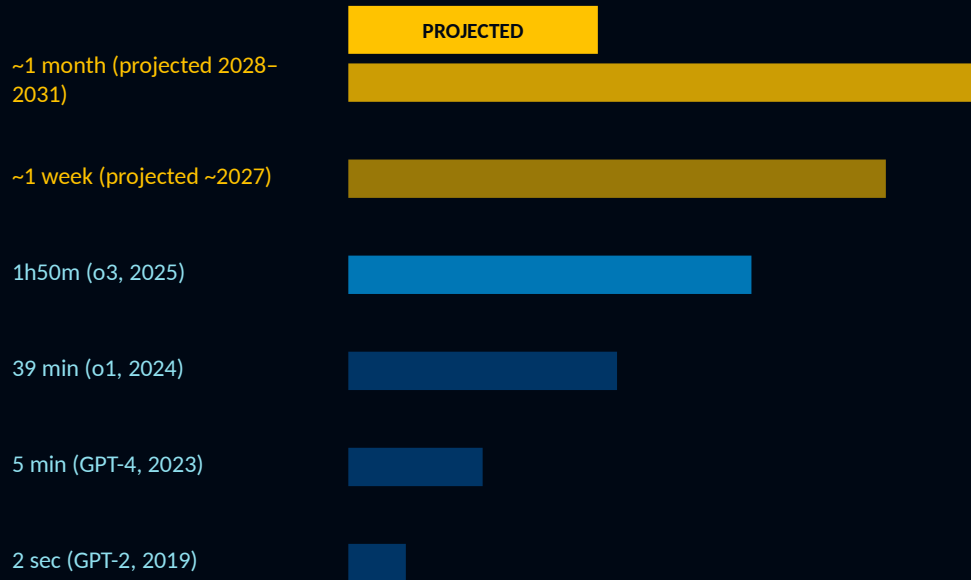
Ability to recognize failures mid-task and adapt strategy without derailing the full execution

LIMITATION

Performance is notably lower on less structured, 'messier' tasks. External validity remains an open question — tasks may not perfectly represent average real-world intellectual labor.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

50% Time Horizon Trend



~1 Month Horizon

by mid-2028 to mid-2031

If current trend continues

~207 days

current doubling time

~20%

faster growth rate 2023–2025 vs 2019–2025

Caveats

Naive extrapolation only. Trend may slow or accelerate.
External validity to real-world jobs uncertain.

Exponential Autonomous Capability Growth Demands Proactive Safety Governance

- 1 **$R^2=0.97$ over 6 years** — 50% time horizon is a robust, intuitive metric for tracking AI autonomous capability growth
- 2 **~7-month doubling time** — possible acceleration since 2024, with 2023–2025 growth ~20% faster than overall trend
- 3 **Frontier at ~2 hours** — o3 can autonomously complete tasks taking skilled humans nearly 2 hours (up from 2 sec for GPT-2)
- 4 **Month-long automation by 2028–2031** — naive extrapolation if 207-day doubling time continues; many software projects automatable
- 5 **Limitations remain** — lower performance on unstructured tasks; external validity to real-world job distributions uncertain

Source: Kwa, West, Becker et al. — METR (2025) | arxiv.org/abs/2503.14499 | 170 tasks, 800+ human baselines